Grokking the Coding Interview: Patterns for Coding Questions

Fast track your coding interview prep by learning coding patterns. Act smartly, and follow the coding patterns.

Level: Intermediate | Study Time: 321h | Lessons: 480 | Playgrounds : 517

211,991 (Students)

Course Overview

Coding interviews are getting harder every day. A few years back, brushing up on key data structures and going through 50-75 coding interview questions was more than enough prep for an interview. Today, everyone has access to massive sets of coding problems, and they have gotten more difficult to account for that. The process has gotten more competitive.

When our team sat together to brainstorm on ideas to make the interview process easier for candidates, we realized quickly that one skill helped us the most when we were preparing for coding interviews: "the ability to map a new problem to an already known problem."

To help candidates with that, we have come up with a list of 27 common coding patterns to help solve coding questions. The idea is, once you are familiar with a pattern, you'll be able to solve dozens of problems with it.

The techniques taught in this course have helped developers land jobs in top companies including Google, Facebook, Amazon, and Microsoft.



Start Reading

Certification on completing the course.

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Course Content

Who Should Take This Course?

Course Overview

2. Warmup — 17 chapters

- Why warmup?
- Contains Duplicate (easy)
- Solution: Contains Duplicate
- Pangram (easy)
- Solution: Pangram
- Reverse Vowels (easy)
- Solution: Reverse Vowels
- Valid Palindrome (easy)
- Solution: Valid Palindrome
- Valid Anagram (easy)
- Solution: Valid Anagram
- Shortest Word Distance (easy)
- Solution: Shortest Word Distance
- Number of Good Pairs (easy)
- Solution: Number of Good Pairs
- Sqrt (medium)
- Solution: Sqrt

3. Pattern: Two Pointers — 23 chapters

- Introduction to Two Pointers Pattern
- Pair with Target Sum (easy)
- Solution: Pair with Target Sum
- Find Non-Duplicate Number Instances (easy)
- Solution: Find Non-Duplicate Number Instances
- Squaring a Sorted Array (easy)
- Solution: Squaring a Sorted Array
- Triplet Sum to Zero (medium)
- Solution: Triplet Sum to Zero
- Triplet Sum Close to Target (medium)
- Solution: Triplet Sum Close to Target
- Triplets with Smaller Sum (medium)

Subarrays with Product Less than a Target (medium)

Solution: Subarrays with Product Less than a Target

Dutch National Flag Problem (medium)

Solution: Dutch National Flag Problem

Problem Challenge 1: Quadruple Sum to Target (medium)

Solution: Problem Challenge 1: Quadruple Sum to Target

Problem Challenge 2: Comparing Strings containing Backspaces (medium)

Solution: Problem Challenge 2: Comparing Strings containing Backspaces

Problem Challenge 3: Minimum Window Sort (medium)

Solution: Problem Challenge 3: Minimum Window Sort

4. Pattern: Fast & Slow Pointers

15 chapters

Introduction to Fast & Slow Pointers Pattern

LinkedList Cycle (easy)

Solution: LinkedList Cycle

Middle of the LinkedList (easy)

Solution: Middle of the LinkedList

Start of LinkedList Cycle (medium)

Solution: Start of LinkedList Cycle

Happy Number (medium)

Solution: Happy Number

Problem Challenge 1: Palindrome LinkedList (medium)

Solution: Problem Challenge 1: Palindrome LinkedList

Problem Challenge 2: Rearrange a LinkedList (medium)

Solution: Problem Challenge 2: Rearrange a LinkedList

Problem Challenge 3: Cycle in a Circular Array (hard)

Solution: Problem Challenge 3: Cycle in a Circular Array

5. Pattern: Sliding Window

21 chapters

Introduction to Sliding Window Pattern

Maximum Sum Subarray of Size K (easy)

Solution: Maximum Sum Subarray of Size K

Smallest Subarray With a Greater Sum (easy)

Solution: Smallest Subarray With a Greater Sum

Longest Substring with K Distinct Characters (medium)

Solution: Longest Substring with K Distinct Characters

Solution: Fruits into Baskets

Longest Substring with Same Letters after Replacement (hard)

Solution: Longest Substring with Same Letters after Replacement

Longest Subarray with Ones after Replacement (hard)

Solution: Longest Subarray with Ones after Replacement

Problem Challenge 1: Permutation in a String (hard)

Solution: Problem Challenge 1: Permutation in a String

Problem Challenge 2: String Anagrams (hard)

Solution: Problem Challenge 2: String Anagrams

Problem Challenge 3: Smallest Window containing Substring (hard)

Solution: Problem Challenge 3: Smallest Window containing Substring

Problem Challenge 4: Words Concatenation (hard)

Solution: Problem Challenge 4: Words Concatenation

6. Pattern: Merge Intervals

15 chapters

Introduction to Merge Intervals Pattern

Merge Intervals (medium)

Solution: Merge Intervals

Insert Interval (medium)

Solution: Insert Interval

Intervals Intersection (medium)

Solution: Intervals Intersection

Conflicting Appointments (medium)

Solution: Conflicting Appointments

Problem Challenge 1: Minimum Meeting Rooms (hard)

Solution: Problem Challenge 1: Minimum Meeting Rooms

Problem Challenge 2: Maximum CPU Load (hard)

Solution: Problem Challenge 2: Maximum CPU Load

Problem Challenge 3: Employee Free Time (hard)

Solution: Problem Challenge 3: Employee Free Time

7. Pattern: Cyclic Sort

17 chapters

Introduction to Cyclic Sort Pattern

Cyclic Sort (easy)

Solution: Cyclic Sort

Find the Missing Number (easy)

Find all Missing Numbers (easy)

Solution: Find all Missing Numbers

Find the Duplicate Number (easy)

Solution: Find the Duplicate Number

Find all Duplicate Numbers (easy)

Solution: Find all Duplicate Numbers

Problem Challenge 1: Find the Corrupt Pair (easy)

Solution: Problem Challenge 1: Find the Corrupt Pair

Problem Challenge 2: Find the Smallest Missing Positive Number (medium)

Solution: Problem Challenge 2: Find the Smallest Missing Positive Number

Problem Challenge 3: Find the First K Missing Positive Numbers (hard)

Solution: Problem Challenge 3: Find the First K Missing Positive Numbers

8. Pattern: In-place Reversal of a Linked List

11 chapters

Introduction to In-place Reversal of a Linked List Pattern

Reverse a LinkedList (easy)

Solution: Reverse a LinkedList

Reverse a Sub-list (medium)

Solution: Reverse a Sub-list

Reverse every K-element Sub-list (medium)

Solution: Reverse every K-element Sub-list

Problem Challenge 1: Reverse alternating K-element Sub-list (medium)

Solution: Problem Challenge 1: Reverse alternating K-element Sub-list

Problem Challenge 2: Rotate a LinkedList (medium)

Solution: Problem Challenge 2: Rotate a LinkedList

9. Pattern: Stacks

16 chapters

Introduction to Stack

Operations on Stack

Implementing Stack Data Structure

Applications of Stack

Problem 1: Balanced Parentheses

Solution: Balanced Parentheses

Problem 2: Reverse a String

Solution: Reverse a String

Problem 3: Decimal to Binary Conversion

Problem 4: Next Greater Element

Solution: Next Greater Element

Problem 5: Sorting a Stack

Solution: Sorting a Stack

Problem 6: Simplify Path

Solution: Simplify Path

10. Pattern: Monotonic Stack (*NEW*)

13 chapters

Introduction to Monotonic Stack

Next Greater Element (easy)

Solution: Next Greater Element

Daily Temperatures (easy)

Solution: Daily Temperatures

Remove Nodes From Linked List (easy)

Solution: Remove Nodes From Linked List

Remove All Adjacent Duplicates In String (easy)

Solution: Remove All Adjacent Duplicates In String

Remove All Adjacent Duplicates in String II (medium)

Solution: Remove All Adjacent Duplicates in String II

Remove K Digits (hard)

Solution: Remove K Digits

11. Pattern: Hash Maps

13 chapters

Introduction to Hashing

Introduction to Hash Tables

Issues with Hash Tables

Problem 1: First Non-repeating Character (easy)

Solution: First Non-repeating Character (easy)

Problem 2: Largest Unique Number (easy)

Solution: Largest Unique Number (easy)

Problem 3: Maximum Number of Balloons (easy)

Solution: Maximum Number of Balloons (easy)

Problem 4: Longest Palindrome(easy)

Solution: Longest Palindrome(easy)

Problem 5: Ransom Note (easy)

Solution: Ransom Note (easy)

- Introduction to Tree Breadth First Search Pattern
- Binary Tree Level Order Traversal (easy)
- Solution: Binary Tree Level Order Traversal
- Reverse Level Order Traversal (easy)
- Solution: Reverse Level Order Traversal
- Zigzag Traversal (medium)
- Solution: Zigzag Traversal
- Level Averages in a Binary Tree (easy)
- Solution: Level Averages in a Binary Tree
- Minimum Depth of a Binary Tree (easy)
- Solution: Minimum Depth of a Binary Tree
- Level Order Successor (easy)
- Solution: Level Order Successor
- Connect Level Order Siblings (medium)
- Solution: Connect Level Order Siblings
- Problem Challenge 1: Connect All Level Order Siblings (medium)
- Solution: Problem Challenge 1: Connect All Level Order Siblings
- Problem Challenge 2: Right View of a Binary Tree (easy)
- Solution: Problem Challenge 2: Right View of a Binary Tree

13. Pattern: Tree Depth First Search

15 chapters

- Introduction to Tree Depth First Search Pattern
- Binary Tree Path Sum (easy)
- Solution: Binary Tree Path Sum
- All Paths for a Sum (medium)
- Solution: All Paths for a Sum
- Sum of Path Numbers (medium)
- Solution: Sum of Path Numbers
- Path With Given Sequence (medium)
- Solution: Path With Given Sequence
- Count Paths for a Sum (medium)
- Solution: Count Paths for a Sum
- Problem Challenge 1: Tree Diameter (medium)
- Solution: Problem Challenge 1: Tree Diameter
- Problem Challenge 2: Path with Maximum Sum (hard)

14. Pattern: Graphs — 11 chapters

- Introduction to Graph
- Graph Representations
- Graph as an Abstract Data Type (ADT)
- Graph Traversal - Depth First Search(DFS)
- Graph Traversal - Breadth First Search (BFS)
- Problem 1: Find if Path Exists in Graph(easy)
- Solution: Find if Path Exists in Graph(easy)
- Problem 2: Number of Provinces (medium)
- Solution: Number of Provinces (medium)
- Problem 3: Minimum Number of Vertices to Reach All Nodes(medium)
- Solution: Minimum Number of Vertices to Reach All Nodes(medium)

15. Pattern: Island (Matrix Traversal) — 15 chapters

- Introduction to Island Pattern
- Number of Islands (easy)
- Solution: Number of Islands
- Biggest Island (easy)
- Solution: Biggest Island
- Flood Fill (easy)
- Solution: Flood Fill
- Number of Closed Islands (easy)
- Solution: Number of Closed Islands
- Problem Challenge 1 (easy)
- Solution: Problem Challenge 1
- Problem Challenge 2 (medium)
- Solution: Problem Challenge 2
- Problem Challenge 3 (medium)
- Solution: Problem Challenge 3

16. Pattern: Two Heaps — 9 chapters

- Introduction to Two Heaps Pattern
- Find the Median of a Number Stream (medium)
- Solution: Find the Median of a Number Stream
- Sliding Window Median (hard)

- Maximize Capital (hard)
- Solution: Maximize Capital
- Problem Challenge 1: Next Interval (hard)
- Solution: Problem Challenge 1: Next Interval

17. Pattern: Subsets

19 chapters

- Introduction to Subsets Pattern
- Subsets (easy)
- Solution: Subsets
- Subsets With Duplicates (easy)
- Solution: Subsets With Duplicates
- Permutations (medium)
- Solution: Permutations
- String Permutations by changing case (medium)
- Solution: String Permutations by changing case
- Balanced Parentheses (hard)
- Solution: Balanced Parentheses
- Unique Generalized Abbreviations (hard)
- Solution: Unique Generalized Abbreviations
- Problem Challenge 1: Evaluate Expression (hard)
- Solution: Problem Challenge 1: Evaluate Expression
- Problem Challenge 2: Structurally Unique Binary Search Trees (hard)
- Solution: Problem Challenge 2: Structurally Unique Binary Search Trees
- Problem Challenge 3: Count of Structurally Unique Binary Search Trees (hard)
- Solution: Problem Challenge 3: Count of Structurally Unique Binary Search Trees

18. Pattern: Modified Binary Search

21 chapters

- Introduction to Modified Binary Search Pattern
- Order-agnostic Binary Search (easy)
- Solution: Order-agnostic Binary Search
- Ceiling of a Number (medium)
- Solution: Ceiling of a Number
- Next Letter (medium)
- Solution: Next Letter
- Number Range (medium)
- Solution: Number Range

Solution: Search in a Sorted Infinite Array

Minimum Difference Element (medium)

Solution: Minimum Difference Element

Bitonic Array Maximum (easy)

Solution: Bitonic Array Maximum

Problem Challenge 1: Search Bitonic Array (medium)

Solution: Problem Challenge 1: Search Bitonic Array

Problem Challenge 2: Search in Rotated Array (medium)

Solution: Problem Challenge 2: Search in Rotated Array

Problem Challenge 3: Rotation Count (medium)

Solution: Problem Challenge 3: Rotation Count

19. Pattern: Bitwise XOR

9 chapters

Introduction to Bitwise XOR Pattern

Single Number (easy)

Solution: Single Number

Two Single Numbers (medium)

Solution: Two Single Numbers

Complement of Base 10 Number (medium)

Solution: Complement of Base 10 Number

Problem Challenge 1: Flip and Invert an Image (hard)

Solution: Problem Challenge 1: Flip and Invert an Image

20. Pattern: Top 'K' Elements

29 chapters

Introduction to Top 'K' Elements Pattern

Top 'K' Numbers (easy)

Solution: Top 'K' Numbers

Kth Smallest Number (easy)

Solution: Kth Smallest Number

'K' Closest Points to the Origin (easy)

Solution: 'K' Closest Points to the Origin

Connect Ropes (easy)

Solution: Connect Ropes

Top 'K' Frequent Numbers (medium)

Solution: Top 'K' Frequent Numbers

Frequency Sort (medium)

Kth Largest Number in a Stream (medium)

Solution: Kth Largest Number in a Stream

'K' Closest Numbers (medium)

Solution: 'K' Closest Numbers

Maximum Distinct Elements (medium)

Solution: Maximum Distinct Elements

Sum of Elements (medium)

Solution: Sum of Elements

Rearrange String (hard)

Solution: Rearrange String

Problem Challenge 1: Rearrange String K Distance Apart (hard)

Solution: Problem Challenge 1: Rearrange String K Distance Apart

Problem Challenge 2: Scheduling Tasks (hard)

Solution: Problem Challenge 2: Scheduling Tasks

Problem Challenge 3: Frequency Stack (hard)

Solution: Problem Challenge 3: Frequency Stack

21. Pattern: K-way Merge

11 chapters

Introduction to K-way Merge Pattern

Merge K Sorted Lists (medium)

Solution: Merge K Sorted Lists

Kth Smallest Number in M Sorted Lists (medium)

Solution: Kth Smallest Number in M Sorted Lists

Kth Smallest Number in a Sorted Matrix (hard)

Solution: Kth Smallest Number in a Sorted Matrix

Smallest Number Range (hard)

Solution: Smallest Number Range

Problem Challenge 1: K Pairs with Largest Sums (hard)

Solution: Problem Challenge 1: K Pairs with Largest Sum

22. Pattern: Greedy Algorithms

13 chapters

Introduction to Greedy Algorithm

Valid Palindrome II (easy)

Solution: Valid Palindrome II

Maximum Length of Pair Chain (medium)

Solution: Maximum Length of Pair Chain

Solution: Minimum Add to Make Parentheses Valid

Remove Duplicate Letters (medium)

Solution: Remove Duplicate Letters

Largest Palindromic Number (Medium)

Solution: Largest Palindromic Number

Removing Minimum and Maximum From Array (medium)

Solution: Removing Minimum and Maximum From Array

23. Pattern: 0/1 Knapsack (Dynamic Programming)

13 chapters

Introduction to 0/1 Knapsack Pattern

0/1 Knapsack (medium)

Solution: 0/1 Knapsack

Equal Subset Sum Partition (medium)

Solution: Equal Subset Sum Partition

Subset Sum (medium)

Solution: Subset Sum

Minimum Subset Sum Difference (hard)

Solution: Minimum Subset Sum Difference

Problem Challenge 1: Count of Subset Sum (hard)

Solution: Problem Challenge 1: Count of Subset Sum

Problem Challenge 2: Target Sum (hard)

Solution: Problem Challenge 2: Target Sum

24. Pattern: Backtracking

11 chapters

Introduction to Backtracking Pattern

Combination Sum (medium)

Solution: Combination Sum

Word Search (medium)

Solution: Word Search

Sudoku Solver (hard)

Solution: Sudoku Solver

Factor Combinations (medium)

Solution: Factor Combinations

Split a String Into the Max Number of Unique Substrings (medium)

Solution: Split a String Into the Max Number of Unique Substrings

- Introduction to Trie
- Implement Trie (Prefix Tree) (medium)
- Solution: Implement Trie (Prefix Tree)
- Index Pairs of a String (easy)
- Solution: Index Pairs of a String
- Design Add and Search Words Data Structure (medium)
- Solution: Design Add and Search Words Data Structure
- Extra Characters in a String (medium)
- Solution: Extra Characters in a String
- Search Suggestions System (medium)
- Solution: Search Suggestions System

26. Pattern: Topological Sort (Graph)

15 chapters

- Introduction to Topological Sort Pattern
- Topological Sort (medium)
- Solution: Topological Sort
- Tasks Scheduling (medium)
- Solution: Tasks Scheduling
- Tasks Scheduling Order (medium)
- Solution: Tasks Scheduling Order
- All Tasks Scheduling Orders (hard)
- Solution: All Tasks Scheduling Orders
- Alien Dictionary (hard)
- Solution: Alien Dictionary
- Problem Challenge 1: Reconstructing a Sequence (hard)
- Solution: Problem Challenge 1: Reconstructing a Sequence
- Problem Challenge 2: Minimum Height Trees (hard)
- Solution: Problem Challenge 2: Minimum Height Trees

27. Pattern: Union Find (*NEW*)

9 chapters

- Introduction to Union Find Pattern
- Redundant Connection (medium)
- Solution: Redundant Connection
- Number of Provinces (medium)
- Solution: Number of Provinces

Solution: Is Graph Bipartite?

Path With Minimum Effort (medium)

Solution: Path With Minimum Effort

28. Pattern: Ordered Set (*NEW*)

9 chapters

Introduction to Ordered Set Pattern

Merge Similar Items (easy)

Solution: Merge Similar Items

132 Pattern (medium)

Solution: 132 Pattern

My Calendar I (medium)

Solution: My Calendar I

Longest Continuous Subarray (medium)

Solution: Longest Continuous Subarray

29. Pattern: Multi-threaded

7 chapters

Introduction to Multi-threaded Pattern

Same Tree (medium)

Solution: Same Tree

Invert Binary Tree (medium)

Solution: Invert Binary Tree

Binary Search Tree Iterator (medium)

Solution: Binary Search Tree Iterator

30. Miscellaneous

2 chapters

Kth Smallest Number (hard)

Solution: Kth Smallest Number

31. Revision

1 chapter

Coding Patterns: A Cheat Sheet

32. Test Your Knowledge (Easy)

11 chapters

Introduction

Two Sum (easy)

Solution: Two Sum

Valid Perfect Square (easy)

Best Time to Buy and Sell (easy)

Solution: Best Time to Buy and Sell Stock

Valid Parentheses (Easy)

Solution: Valid Parentheses

Subtree of Another Tree (easy)

Solution: Subtree of Another Tree

33. Test Your Knowledge (Medium)

52 chapters

Daily Temperatures (Medium)

Solution: Daily Temperatures

Group Anagrams (Medium)

Solution: Group Anagrams

Decode String (Medium)

Solution: Decode String

Valid Sudoku (Medium)

Solution: Valid Sudoku

Product of Array Except Self (Medium)

Solution: Product of Array Except Self

Maximum Product Subarray (Medium)

Solution: Maximum Product Subarray

Container With Most Water (Medium)

Solution: Container With Most Water

Palindromic Substrings (Medium)

Solution: Palindromic Substrings

Remove Nth Node From End of List (Medium)

Solution: Remove Nth Node From End of List

Find Minimum in Rotated Sorted Array (Medium)

Solution: Find Minimum in Rotated Sorted Array

Pacific Atlantic Water Flow (Medium)

Solution: Pacific Atlantic Water Flow

Validate Binary Search Tree (medium)

Solution: Validate Binary Search Tree

Construct Binary Tree from Preorder and Inorder Traversal (medium)

Solution: Construct Binary Tree from Preorder and Inorder Traversal

Clone Graph (medium)

House Robber II (medium)

Solution: House Robber II

Decode Ways (medium)

Solution: Decode Ways

Unique Paths (medium)

Solution: Unique Paths

Word Break (medium)

Solution: Word Break

Lowest Common Ancestor of a Binary Search Tree (medium)

Solution: Lowest Common Ancestor of a Binary Search Tree

Longest Consecutive Sequence (medium)

Solution: Longest Consecutive Sequence

Meeting Rooms II (medium)

Solution: Meeting Rooms II

Encode and Decode Strings

Solution: Encode and Decode Strings

Number of Connected Components in an Undirected Graph

Solution: Number of Connected Components in an Undirected Graph

Graph Valid Tree (medium)

Solution: Graph Valid Tree

Implement Trie (Prefix Tree) (medium)

Solution: Implement Trie (Prefix Tree)

Design Add and Search Words Data Structure (medium)

Solution: Design Add and Search Words Data Structure

34. Test Your Knowledge (Hard)

4 chapters

Longest Valid Parentheses (hard)

Solution: Longest Valid Parentheses

Serialize and Deserialize Binary Tree (hard)

Solution: Serialize and Deserialize Binary Tree

35. Conclusions

1 chapter

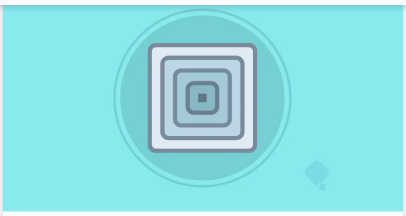
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Algorithms can be daunting, but they're less so with the right guide. This course - https://www.designgurus.io/course/grokking-the-coding-interview, is a great starting point. It covers typical problems you might encounter in interviews.

MO JAFRI
Software Engineer

The courses which have "grokking" before them, are exceptionally well put together! These courses magically condense 3 years of CS in short bite-size courses and lectures (I have tried System Design, OODI, and Coding patterns). The Grokking courses are godsent, to be honest.

AHMET HANIF
Software Engineer

Whoever put this together, you folks are life savers. Thank you :)

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